ATM Kinase Activation by DNA Double-Strand Breaks Induces p53 Phosphorylation and p21-Mediated Arrest

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Page 1 — Abstract

Genomic integrity is maintained by complex DNA damage response surveillance networks. Double-strand breaks rapidly activate the apical sensor kinase ATM.

Page 2 — Introduction

Activated ATM kinase directly phosphorylates p53 at Ser15, preventing MDM2-mediated ubiquitination and degradation.

Page 3 — Methods & Assays

Stabilized nuclear p53 binds canonical promoter response elements to induce p21 transcription in irradiated fibroblasts.

Page 4 — Results

Elevated p21 protein binds and inhibits cyclin-dependent kinase 2 (CDK2) complexes, preventing retinoblastoma phosphorylation.

Page 5 — Discussion

Through this axis, ATM kinase activation induces p21 expression via p53 to enforce cell cycle arrest at the G1/S checkpoint.